Aerodynamic Optimization of Space Plane Control Surfaces

Optimizing Aerodynamics and Control Surfaces of a Blended Lifting-Body Canard-Delta Wing Space Plane

---

Introduction

The development of space planes leveraging blended lifting-body and canard-delta wing configurations represents the vanguard of aerospace engineering, marrying the aerodynamic refinement necessary for efficient atmospheric flight with the robust maneuverability and stability essential for safe orbital transitions and survivability during hypersonic reentry. This report provides a comprehensive synthesis of modern techniques, design principles, simulation tools, and integrated multidisciplinary optimization methodologies that underpin the performance enhancements in such vehicles. Emphasis is placed on the interplay between aerodynamic shaping, control surface allocation, systems integration (including propulsion and thermal protection), and advanced simulation-driven optimization, supported by recent high-fidelity, multidisciplinary research and real-world case studies.

Aerodynamic Design Principles of Blended Lifting-Body Configurations

The **blended lifting-body (BLB)** and **blended wing-body (BWB)** architectures revolutionize space plane design by integrating the fuselage, wings, and sometimes propulsion, into a continuous, low-drag, all-lifting surface. This continuity reduces wet area, mitigates interference drag, and increases structural efficiency due to distributed load paths and mass centralization[2][4]. The elimination of discrete fuselage-wing junctions characteristic of conventional tube-and-wing designs lowers skin-friction losses and creates a platform well-suited for high lift-to-drag (L/D) ratios, a key metric for efficiency in both subsonic and supersonic regimes[6][7].
In these configurations, the thick centerbody not only serves as a payload bay and supports embedded propulsion but also carries much of the lifting load. Studies show that optimized lifting-bodies can achieve up to 27-36% reductions in overall fuel burn and empty weight relative to conventional aircraft, largely driven by reductions in parasitic and induced drag[8]. The careful volumetric shaping, akin to Sears-Haack or Whitcomb's area-ruled bodies, ensures favorable pressure recoveries at transonic and supersonic speeds, reducing wave drag that would otherwise be prohibitive for high-speed atmospheric or reentry flight[5].
Reflex cambered airfoils and planform features tailored for tailless stability-such as positive dihedral, sweep, and distributed twist-are common, as they provide the required negative

pitching moments absent from discrete horizontal tails, supporting passive or active trim in cruise[9]. Lateral stability, sometimes a weakness in tailless designs, is mitigated through proper wing sweep, dihedral, and in some cases, wingtip fins or small vertical stabilizers, without incurring major drag penalties[2].

Aerodynamic Characteristics of Canard-Delta Wing Integration

The integration of **canard foreplanes** with delta main wings further improves the maneuverability envelope and static/dynamic stability, especially during transonic ascent/descent and in the upper atmosphere[11][8]. Canards-mounted forward of the main wing-generate positive lift and favorable pitching moments, shifting the aerodynamic center forward and enhancing pitch control authority. At high angles of attack, canard-generated vortices augment lift on the main wing through controlled vortex interaction and delayed flow separation, increasing the maximum achievable lift coefficient and expanding the safe operational AoA envelope[11][8].
The **close-coupling** of canard and main wing is critical: optimal elevation and proximity enhance beneficial vortex interaction, yielding increased L/D, higher stall angles, and improved post-stall controllability. However, integration also introduces new complexity, such as balancing the increased pitch-up moment against the need to maintain sufficient static stability, especially when propulsive or payload mass distributions shift the center of gravity[11][8].
At supersonic speeds, the aerodynamic center of the entire vehicle tends to shift aft, reducing inherent static stability. Canard surfaces-particularly those with variable geometry or large movable ranges-can be used to tailor pitch authority to counteract such tendencies and reduce trim drag, thus supporting efficient high-Mach cruise and reentry[8].

Key Aerodynamic Features, Control Surface Types, and Functions

Aerodynamic Feature	Control Surface Type	Function
Blended Lifting-Body	Trailing edge elevons, body flaps	Multi-axial pitch/roll control, trim authority, and lift enhancement
Canard Foreplane	All-moving canard	Pitch trim, lift augmentation, static/dynamic stability enhancement
Delta Main Wing	Leading/trailing edge flaps	High-lift device, roll/yaw manipulation, stall delay
Wingtip Fins/V-Stabs	Rudder, ruddervators	Yaw control, lateral stabilization
Split Drag Rudders	Clamshell elevons	Differential yaw/drag control, speedbrake function
Thermal Leading-Edge Panels	TPS-integrated flaperons	Aerodynamic control with thermal protection during reentry
Redundant Actuators	Dual/Triple redundant systems	Maintain control authority after actuator failure

Integrated Propulsion Control	Thrust vectoring vanes	Coupled pitch/yaw control during ascent and reentry

Each of these features is typically optimized and validated via multi-regime simulation and wind tunnel testing[12][13].

Control Surface Design and Placement on Blended Lifting-Body Space Planes

Principles and Methods

In the absence of traditional tails in BWB/BLCB designs, **control surfaces** must assume multi-functional roles. *Elevons*-movable surfaces on the trailing edge of the wing-enable simultaneous pitch and roll control; *body flaps* provide high-authority control at high angles of attack, such as during hypersonic descent; *canard surfaces* deliver forward pitch authority and supplement overall lift, especially during lower-speed and high-maneuvering phases. The placement and sizing of these surfaces are pivotal, as their proximity to the vehicle's center of gravity and aerodynamic center directly dictate control effectiveness and structural loads[14][16].
Optimal allocation is generally governed by the following design principles:

- Surfaces are distributed so that their combined effectiveness covers all three axes (pitch, yaw, roll) under a range of center-of-gravity positions and flight conditions.
- Surface areas are balanced to avoid excessive control power requirements (which would increase actuator size and risk flutter).
- Redundant multi-surface configurations are designed to provide fault tolerance and ensure continued control authority after actuator/surface failure.
- Inter-surface interference is minimized by avoiding large, simultaneous adjacent deflections.

Advanced control effectors such as **split drag rudders**, **multi-modal elevons**, and **flaperons** (fusing aileron and flap functions) have been demonstrated in X-48 and HL-20 prototypes and wind tunnel studies, revealing robust control effectiveness with proper allocation and minimal cross-coupling effects under most conditions[4][9].

Static Stability Optimization Strategies for Lifting-Body Vehicles

The optimization of **static stability** in all-lifting configurations depends heavily on the combination of airfoil selection, control surface authority, and center-of-gravity control:

- **Airfoil Reflex Design**: Reflex cambered or "S-shaped" airfoils provide an inherent nose-down pitching moment counterbalancing lift forces that would otherwise promote pitch-up instability. These are widely used in tailless and BWB vehicles[3].
- **Canard Integration**: Canards are sized so that their stall angle is reached before the main

wing, conferring an “anti-stall” condition that ensures the vehicle nose is forced down prior to main wing stall, preserving controllability near the boundary[8].

- **Elliptical Lift Distribution and CG Placement**: Careful alignment of center-of-gravity and the mean aerodynamic chord produces favorable static margins (typically 5-15% for robust flight). Static margin is checked via simulation across fuel/payload mass variations to ensure safety during all mission phases[3].

Robustness against off-nominal conditions is improved through **multipoint optimization**-where the airframe is optimized across a range of flight regimes (subsonic, transonic, supersonic, and hypersonic)-as opposed to only during cruise, yielding solutions with high tolerance for atmospheric disturbances and trajectory deviations.

Dynamic Stability and Maneuverability Trade-Offs

Dynamic stability in space planes is characterized by the fast response of short-period (oscillatory) motions, phugoid (long-period) motions, Dutch roll, spiral, and roll mode dynamics. The lack of a horizontal tail in BWB/BLCB platforms shifts responsibility for these to the elevons, canards, and associated high-authority control surfaces.
Critical aspects include:

- **Control surface effectiveness must be maintained across the dynamic envelope**, particularly at high-deflection angles where nonlinearities rapidly reduce effectiveness due to local flow separation[7].
- **Active fly-by-wire systems** are essential for exploiting relaxed static stability configurations, allowing for smaller surfaces, reduced weight, and minimized drag, while maintaining required handing qualities through continuous dynamic control correction[3].
- **Yaw, roll, and pitch authority are independently optimized** through sophisticated control allocation algorithms that assign requested moments to available surfaces, balancing effectiveness, actuator loads, and minimization of drag[14].
- **Canard/Delta interaction** augments maneuverability without unduly penalizing static or dynamic stability. However, special care is needed to avoid adverse yaw and undesirable interactions with main wing induced flow, especially near stall conditions[8].

The trade-off is often between absorbing greater control complexity (in both surface design and flight control computation) in exchange for extraordinary flight envelope expansion, weight reduction, and efficiency gains[15].

Multi-Disciplinary Design Optimization in Aerospace Engineering

Multi-disciplinary design optimization (MDO) is the central paradigm enabling simultaneous, system-level improvement of aerodynamics, structure, propulsion, control, and thermal management. Modern MDO frameworks (e.g., OpenMDAO, MACH, ISIGHT, OPTIMUS, SUAVE)

integrate diverse analysis codes-CFD, FEA, thermal solvers, propulsion cycle models-into closed-loop optimization algorithms[18][20][21].
Key features include:

- **Adjoint and gradient-based optimization**: Exploit efficient derivative calculation for hundreds of shape or structural parameters, enabling rapid convergence toward minima in drag, weight, or a multi-objective cost function[23][24].
- **Surrogate and reduced-order modeling**: Neural networks, proper orthogonal decomposition (POD), and kriging models are layered atop high-fidelity simulations to accelerate design exploration, especially where computational cost is prohibitive[19].
- **Integration of constraints**: Constraints such as static/dynamic stability margins, actuator sizing, thermal protection thickness, and mission-specific trajectory performance are enforced during the optimization process, ensuring realistic and certifiable solutions[19].
- **Uncertainty quantification and robustness analysis**: Advanced techniques quantify how design variations, material property uncertainties, or operational variability influence performance, securing operational reliability[18].

Frameworks have matured to the point that entire aircraft, including blended-wing-body and space plane variants, can be optimized through mission profiles, adjusting airframe, surface, actuator, and system parameters in a coordinated fashion[25].

Adjoint and Gradient-Based Aerodynamic Shape Optimization Techniques

Adjoint Methods in Aerodynamics

The **adjoint method** underlies most high-efficiency aerodynamic optimizations, allowing for rapid calculation of derivatives of objective functions (e.g., drag, lift, pressure loss) with respect to all design variables, independent of the number of variables. This property is invaluable in optimizing complex geometries such as blended lifting-bodies and canard-delta surfaces, where hundreds of control points may be actively varied[23][24].

Workflow

1. **Baseline CFD Simulation:** High-fidelity solver (RANS, URANS) provides the flow field for a given geometry.
2. **Adjoint Problem Solution:** Computes the sensitivities of the chosen objective function to local geometry changes (surface node displacements).
3. **Shape Update:** Deformation vectors, guided by adjoint sensitivities, drive mesh morphing (often via free-form deformation or direct mesh movement), yielding a new candidate geometry.
4. **Iterative Optimization:** The process repeats for multiple objectives/constraints until convergence to an optimal solution.

Gradient-based (especially adjoint-driven) methods are critical for handling **multi-objective** (e.g., maximizing L/D while minimizing heating) and **multi-condition** (e.g., performance at multiple Mach numbers/AoA) problems. These methods are implemented in commercial and research codes such as Ansys Fluent, DLR TAU, SUAVE, CAESES, ADflow, and CODA[24][19].

CFD and Simulation Tools for Aero-Propulsive Optimization

Computational Fluid Dynamics (CFD) and simulation tools are the backbone of modern space plane optimization, offering both high-fidelity prediction and rapid design iteration. Key characteristics include:

- **Reynolds-Averaged Navier-Stokes (RANS) Solvers:** Used for steady and unsteady high-speed flow, viscous drag, heat flux, and separated regions (e.g., DLR-TAU, ADflow, ANSYS Fluent, CODA)[19][8].
- **Aeroelastic and Fluid-Structure Interaction (FSI):** Coupled CFD and structural solvers are essential for assessing control surface deflection under aero loads, modeling flutter, and predicting buzz instabilities[27][28].
- **Mesh Handling:** Techniques such as mesh morphing via Radial Basis Functions, Chimera (overset) grid methods, and sliding interfaces allow resolving large control surface deflections, dynamic motion, and highly swept delta geometries[26].
- **Integration with Propulsion Modeling:** OpenMDAO, pyCycle, and similar tools support coupled aeropropulsive simulation where inlet flow, fan pressure ratios, and heat loads directly influence aerodynamic shaping and vice versa[20][29].

Specialized solvers, such as ADflow and CODA, enable analytic adjoint derivation for gradient-based MDO. Their suitability has been demonstrated for single-point and multipoint optimizations, including boundary layer ingestion and distributed propulsion cases relevant for future hybrid-electric and hydrogen-powered space planes[20][29].

Aeroelastic and Fluid-Structure Interaction Modeling for Control Surfaces

Aerodynamic loading on control surfaces induces local deformations (especially in large, high-lift surfaces or flexible structures) that, in turn, feedback into the aerodynamic solution-a phenomenon critical in **aeroelastic modeling**. Robust prediction underlies the prevention of **flutter, buzz instability, and loss of control authority**.

- **Coupled CFD-CSM Solvers:** RANS-based CFD is linked with modal finite element analyses (FEM), iteratively transferring pressure loads and structural deformations until convergence [27][7].
- **Overset Meshes (Chimera):** Allow for efficient simulation of moving control surfaces without regenerating the mesh for every deflection angle, saving time and preserving accuracy[26].

- **Sliding Interfaces:** Enable continuity across mesh blocks with moving surfaces, validated for surface deflections up to and beyond ±30°, supporting both static and transient (flutter/buzz) analyses[27].

Wind tunnel and full-vehicle simulations demonstrate that accurate aeroelastic modeling is vital for ensuring the integrity and effectiveness of all-movable control surfaces at high speeds and under reentry thermal loads[7].

Control Allocation and Effectiveness Modeling for Multifunctional Control Surfaces

Control allocation is the formalism by which pilot or autopilot commands (moment/force requests about various axes) are translated into actuator demands for each control surface. For redundant, coupled, and multi-axis surfaces typical in BWB and canard-delta designs, this becomes a multidimensional optimization problem[15][9].

Key findings and modern practices:

- **Linear and Nonlinear Allocation:** Algorithms typically initiate with linear control effectiveness matrices, but nonlinearities due to large deflections, stall, and flow separation critically degrade authority if not modeled accurately[15].
- **Optimal Allocation Objectives:** Trade-offs between minimal control effort, minimal induced drag, real-time computational feasibility, and structural load balancing guide algorithm selection. Formulations range from weighted least-squares and pseudo-inverse (linear), to l1/l2/l∞ norm-based optimizations and dynamic inversion feedback (nonlinear)[15].
- **AMS (Attainable Moment Set):** Used to quantify the achievable control authority margin, informing system-level design and surface sizing[9].
- **Fault-Tolerant Operation (Redundancy):** Allocation logic anticipates possible failures, ensuring sufficient moment authority remains in degradated system modes[31][32].

Practically, these allocations have been validated on prototypes such as NASA’s X-48B, where pitch, roll, and yaw authority proved sufficient even in the presence of simultaneous actuator failures[31].

Integration of Propulsion Systems with Aerodynamic Design

The synergy between **aerodynamic shaping** and **propulsion system integration** is a driving factor in next-generation space plane development:

- **Boundary Layer Ingestion (BLI):** Integrating engines to ingest low-momentum boundary layer flow reduces wake energy loss, potentially lowering required power up to 10%. BWB configurations with upper-surface engines exemplify this approach[20][29].
- **Distributed Electric Propulsion:** Electric/hybrid propulsion architectures facilitate distributed, embedded thrust along the body, improving spanwise load distribution and

control authority. Coupling with advanced CFD and propulsive cycle models, such as those using ADflow/pyCycle frameworks, enables full performance mapping and optimization[29].

- **Thermal Integration:** Thermal loads from embedded engines, reentry friction, and TPS interaction are modeled simultaneously to ensure survivable structures and effective heat shielding[34][2].
- **Multi-point Aeropropulsive Optimization:** MDO approaches treat cruise and off-design points (e.g., rolling-takeoff) as joint constraints, ensuring that design adjustments (like increasing nacelle diameter) benefit overall flow uniformity and maintain propulsion efficiency across the mission envelope[20][29].

Analysis across many studies supports the necessity for robust, gradient-driven optimization that includes fan thrust, pressure ratios, and heat loads as variables in airframe optimization[20][29].

Thermal Protection System Aerodynamic Integration

Thermal protection systems (TPS) must balance aerodynamic efficiency with survivability during reentry:

- **Tile and Insulation Sizing:** Advanced simulation tools (thermal FEA and trajectory-based heating loads) enable regional TPS sizing optimized for weight and thermal performance, minimizing overdesign and excessive mass penalty[34].
- **Tile Bond Verification:** Automated, instrumented force sensors and DAQ systems ensure surface conformity and bond strength for TPS tiles covering complex, curved blended surfaces, as seen in Dream Chaser and Shuttle-derived designs[34].
- **Aerodynamic Integration:** Shaping for aerothermal loads is performed early in MDO cycles, with dedicated regions for high, medium, and low TPS capability. Placement of instrumentation, control surfaces, and structural joints is optimized to minimize flow separation and TPS breach risk under hypersonic flow[34][33].

The approach is increasingly multidisciplinary, simultaneously considering aerodynamic pressures, thermal flux, surface integrity, actuator routing, and maintenance accessibility[33].

Case Studies of Blended-Wing-Body and Space Plane Prototypes

Numerous prototypes have validated these principles in hardware and simulation:

- **NASA HL-20**: Extensive wind tunnel programs assessed elevon trim authority, body flap roll/yaw control, and integrated aerodynamic-TPS interaction. Results indicated robust static and dynamic stability, with skewed hinge body flaps mitigating adverse yaw, and yaw controllers providing favorable rolling moment for lateral control during hypersonic descent [13][36].
- **Boeing X-48B/C and Airbus BWB:** Demonstrated multi-surface elevon/rudderon layout, high

passive stability via reflex airfoils and sweep, and control allocation resilience to actuator/surface failures[9].

- **Dream Chaser® Spaceplane:** Embodied integrated design of aerodynamics, propulsion, and TPS tiles, with vastly reduced number of larger, more robust tiles (compared to Shuttle), and automated verification of aerodynamic fit and bond strength, increasing operational reliability and turn-around time for reuse[34].
- **Reusable Launch and Reentry Vehicles (RLVs):** Modern concepts-Skylon, IXV, HYFLEX-continue to refine control surface placement, propulsion integration with boundary layer ingestion, and TPS/structure thermal optimization[6][37].

Lessons from these case studies confirm the practicality and necessity of multidisciplinary, simulation-driven design, balanced with ground and flight testing to validate prediction.

Control Surface Redundancy and Failure Mode Analysis in Space Plane Control Systems

Given the reliance of unstable or tailless configurations on control surfaces for stability, rigorous **redundancy and failure management** is essential:

- **Multiple independent actuators per surface**: Three- or four-channel actuation, with force-voting or active/standby switching, is mandated for critical control lines in flight phases such as boost, reentry, and landing[31][32].
- **Redundant computational chains**: Flight-critical computer systems operate in synchronized, redundant fashion, with comparator and voter logic to detect, isolate, and reassign tasks on failure[31].
- **Simulation and validation**: Simulation tools such as NASA's FSAA and modeled redundancy management logic verify operational robustness under single/multi-point failures and provide insight for certification[32].
- **Fail-operational logic**: After a single actuator/surface failure, remaining units must sustain control envelope, possibly with degraded performance but within safe limits-demonstrated via actuation simulation, common cause failure modeling, and reliability analysis[32].

This robust approach ensures survivability and flight safety even in complex, integrated BWB/canard-delta architectures.

Emerging Trends and Key Players in Space Plane Aerodynamic Optimization

Technological Trends

- **Morphing Wings and Adaptive Control Surfaces**: The rapid maturation of **morphing technologies**-electromechanically actuated flexible structures capable of in-flight shape changes-enables aircraft to adapt aerodynamic properties actively for optimal efficiency and handling throughout the flight envelope[28].

- **Digital Twin and Data-Driven Design**: Extensive use of digital twins, AI-driven optimization, and real-time data collection facilitates not only advanced design and predictive maintenance but also validation and certification, critical as control systems and surface complexity grow [40].
- **Electrified/Hybrid Propulsion**: Integration of distributed and hybrid-electric propulsion, hydrogen fuel cells, and novel boundary layer ingestion engines continues, as environmental and efficiency pressures increase, requiring even tighter coupling between propulsion and aerodynamic optimization[39][40].

Industry and Research Players

- **NASA, Boeing, and Airbus**: Leading extensive research in BWB and lifting-body concepts, flying demonstrators, and pioneering aerodynamic/propulsive integration strategies.
- **European and Asian R&D**: Airbus, ESA, JAXA, and Chinese institutes push boundaries on canard delta and reentry vehicle design, with IXV, HYFLEX, and next-generation hypersonic studies.
- **DLR (German Aerospace Center)**, **ONERA**, and leading universities provide tool (DLR-TAU, CODA), algorithm, and methodology development for multi-fidelity, multi-disciplinary simulation and optimization[26][9].

Conclusions and Design Recommendations

Optimizing the aerodynamics and control surfaces of a blended lifting-body, canard-delta wing space plane requires synthesis of advanced aerodynamic shaping, robust control surface design, integrated propulsion/TPS strategies, and multi-disciplinary, simulation-driven optimization frameworks.
Design recommendations:

- Use blended lifting-body planforms and reflex airfoils for superior baseline stability and lift/drag performance.
- Integrate canard-foreplanes for dynamic pitch control, extended maneuverability, and expanded stall boundaries; closely couple canard-vortex interaction with main wing performance.
- Distribute multi-functional, redundant control surfaces (elevons, flaperons, body flaps) using allocation algorithms validated in both linear and nonlinear response regimes.
- Rely on gradient-based and adjoint optimization methods for high-dimensional, multi-objective shape and systems optimization.
- Integrate propulsion modeling and TPS analysis early and through coupled simulation for realistic trajectory, load, and thermal predictions.

- Adopt morphing and adaptive technologies where possible to maximize efficiency across diverse flight conditions.
- Validate control effectiveness, redundancy, and survivability through aggressive wind tunnel, simulation, and eventually flight test, including failure mode and recovery analysis.

The future of space plane design sits at the intersection of robust multidisciplinary simulation, reusable system architecture, and adaptive, intelligent control systems-a synthesis best exemplified by the ongoing research and industry collaboration in blended lifting-body, canard-delta wing vehicles.

Aerodynamic Features, Control Surface Types, and Functions Summary Table

Aerodynamic Feature	Control Surface Type	Functionality
Blended Lifting-Body	Elevons, body flaps	Pitch & roll control, cruise trim
Canard-Delta Wing	Forward canards	Pitch trim, maneuverability, static/dynamic stability enhancement
Trailing Edge (Delta)	Flaperons, split drag rudder	Pitch, roll, yaw, speedbrake, and redundancy
TPS Integrated Surfaces	Thermal flaps	Aerothermodynamic integrity during reentry
Embedded Propulsion	Thrust vectoring vanes	High-authority pitch & yaw, reentry and ascent control
Surrogate Shape Functions	Morphing surfaces	Adaptive shaping for efficiency and envelope expansion
Redundant Actuation	Multiple actuators	Ensures fail-operational control after surface or system failure

By adhering to these practices, future blended lifting-body, canard-delta wing space planes can achieve unparalleled performance in efficiency, maneuverability, and safety across all phases of atmospheric and orbital flight.

References (40)

1. *Multidisciplinary Design Optimization Processes for Efficiency ...*. https://link.springer.com/article/10.1007/s42405-024-00811-8
2. *Improving Safety Condition in Construction Projects in Iran*. https://www.akademiabaru.com/doc/ARFMTSV53_N2_P165_174.pdf
3. *Development of the DLR TAU Code for Modelling of Control Surfaces - DGLR*. https://www.dglr.de/publikationen/2019/480018.pdf

4. *Aerodynamic Effectiveness Configuration Characteristics of the*. https://ntrs.nasa.gov/api/citations/19990117251/downloads/19990117251.pdf
5. *Title presentation - HAW Hamburg*. https://www.fzt.haw-hamburg.de/pers/Scholz/ewade/2014/05_Innovative_Concepts_S2_C_TUD.pdf
6. *Design, Control Surface Optimization and Stability Analysis of a …*. https://www.sae.org/publications/technical-papers/content/2021-01-0040/
7. *Control allocation performance for blended wing body aircraft and its …*. https://daneshyari.com/article/preview/10681290.pdf
8. *Microsoft Word - IJIEEB-V2-N1-6-ICIECS2010-4114.doc - MECS Press*. https://www.mecs-press.org/ijieeb/ijieeb-v2-n1/IJIEEB-V2-N1-6.pdf
9. *Preliminary Subsonic Aerodynamic Model for Simulation Studies of the HL …*. https://ntrs.nasa.gov/api/citations/19920021931/downloads/19920021931.pdf
10. *Aerodynamic Design and Exploration of a Blended Wing Body Aircraft at …*. https://commons.erau.edu/ijaaa/vol6/iss5/17/
11. *Best Design of Wing Aerodynamic Control Surfaces Based on … - Springer*. https://link.springer.com/article/10.1007/s42417-024-01458-1
12. *Influence of Canard on the Longitudinal and Lateral-Directional …*. https://www.iccfd.org/iccfd11/assets/pdf/papers/ICCFD11_Paper-2005.pdf
13. *Template For The Preparation Of Papers For On-Line Publishing In ODE - ICAS*. https://www.icas.org/icas_archive/ICAS2012/PAPERS/112.PDF
14. *Tools and Challenges in Evaluating Control Surface Airworthiness for a …*. https://kth.diva-portal.org/smash/get/diva2:1817055/FULLTEXT01.pdf
15. *Design and Analysis of Blended Wing Body Aircraft for Stability*. https://link.springer.com/chapter/10.1007/978-981-19-6945-4_66
16. *Coupled Aeropropulsive Design Optimization of a Boundary Layer …*. https://openmdao.org/pubs/gray_bli_opt_2018.pdf
17. *Coupled aeropropulsive design optimization of a podded electric …*. https://link.springer.com/article/10.1007/s00158-024-03904-w
18. *Shape Optimisation without constraints - How to use the Adjoint Solver …*. https://www.leapaust.com.au/blog/cfd/shape-optimisation-adjoint-solver-part-1/
19. *Shape Optimization with Adjoint CFD - CAESES*. https://www.caeses.com/products/caeses/adjoint-cfd-shape-optimization/
20. *AircraftsimulationsusingthenewCFDSoftware fromONERA,DLR,andAirbus*. https://hal.science/hal-04575297/document
21. *Bombardier standard presentation for Aerospace and Transportation*. https://www.fzt.haw-hamburg.de/pers/Scholz/ewade/2015/SCAD2015_Piperni_MDOforAircraftDesignAtBombardier.pdf
22. *Aeroelastic analysis of aircraft with control surfaces using CFD*. https://theses.gla.ac.uk/5499/
23. *Aeroelastic Modeling of Elastically Shaped Aircraft Concept via Wing …*. https://ntrs.nasa.gov/api/citations/20170005442/downloads/20170005442.pdf
24. *TAU - dlr.de*. https://www.dlr.de/en/as/research-and-transfer/software-solutions/aerodynamics/software-tau

25. *Advancements in Coupled Aeropropulsive Design Optimization*. https://www.nas.nasa.gov/pubs/ams/2025/05-01-25.html
26. *Redundancy Management Technique Space Shuttle Computers*. https://people.cs.rutgers.edu/~uli/cs673/papers/RedundancyManagementSpaceShuttleIBM76.pdf
27. *PowerPoint Presentation*. https://ntrs.nasa.gov/api/citations/20170012470/downloads/20170012470.pdf
28. *Spaceplane Thermal Protection System Control* . https://dewesoft.com/blog/space-thermal-protection-system-fitting-control
29. *Thermal Protection System Analysis and Sizing for Spaceplane ...*. https://repository.tudelft.nl/record/uuid:dbd513c7-346d-4a7b-972c-7eb9497f1bd8
30. *NASA HL-20 Lifting Body Airframe - MATLAB & Simulink - MathWorks*. https://www.mathworks.com/help/aeroblks/nasa-hl-20-lifting-body-airframe.html
31. *Aerodynamic Configuration of the HCW Based on the Lifting Body*. https://www.sciencegate.app/document/10.1061/%28asce%29as.1943-5525.0000950
32. *A Transformative Year in Aerospace - Aerospace America*. https://aerospaceamerica.aiaa.org/departments/a-transformative-year-in-aerospace/
33. *Future-Ready Aircraft 2025 - taaltech.com*. https://www.taaltech.com/future-ready-aircraft-trends-shaping-aerospace-design-in-2025-and-beyond/